Annotating and exploring Medieval Nordic texts

Paul Meurer

Universitetet i Bergen, UB

10 September 2026

Innhold

1. Overview: Menota and the Korpuskel application
2. Korpuskel as an editorial tool
3. Korpuskel (and INESS) as tools for linguists
4. The Menota Lemmatizing Assistant (MLA)

Content

1. Overview: Menota and the Korpuskel application
2. Korpuskel as an editorial tool
3. Korpuskel (and INESS) as tools for linguists
4. The Menota Lemmatizing Assistant (MLA)

Korpuskel: Requirements

Menota needs a web-based, versatile tool that supports the full range of features menotic texts offer.

Two feature sets: editorial and linguistic

Requirements for editorial features

- Parallel representation of texts on three levels (facsimile, diplomatic, normalized)
- Styling of textcritical annotations (according to Menota Handbook guidelines)
- Pagination
- Display of facsimile images
- Display of aligned translations
- Linking to dictionary entries

Korpuskel: Requirements

Requirements for linguistic features

- Display of morphological annotations
- Display of syntactical annotations
- Corpus functionality, e.g. querying, KWIC concordances, collocations, distribution statistics
- Search in syntactic annotations (cf. the Menotec and INESS projects)
- MLA (Menota Lemmatizing Assistant): tool to do semi-automatic annotations at norm and facs level, as well as annotation of lemma, part of speech and morphological features

A bit of history: Menota, Korpuskel and the MLA

- 2001: Menota corpus tool based on Corpus Workbench
- 2006: MLA (Menota Lemmatizing Assistant), first version
- 2010: Korpuskel: Project proposal for the development of a new versatile corpus engine and web interface (Meltzer foundation)
- 2016: New, Korpuskel-based Menota corpus tool
- 2023: Better user interface based on modern UI development tools, focus on Universal design and accessibility (Clarino)
- 2023: MLA 2.0

Project collaboration

Project collaboration

Odd Einar:
Subject: Verre enn nokon gong!

Hallo Paul,

No verkar ikkje konkordanssøket i det heile teke - det kjem berre opp blanke linjer. Kan du rette dette opp no? Vi sit og arbeider med tekstane i dag, og dette er svært keisamt.

MVH Odd Einar

Project collaboration

Odd Einar:
Subject: Verre enn nokon gong!

Hallo Paul,

No verkar ikkje konkordanssøket i det heile teke - det kjem berre opp blanke linjer. Kan du rette dette opp no? Vi sit og arbeider med tekstane i dag, og dette er svært keisamt.

MVH Odd Einar

Odd Einar:
Subject: Panikk! I alle fall litt …
...

Content

1 Overview: Menota and the Korpuskel application

2 Korpuskel as an editorial tool

3 Korpuskel (and INESS) as tools for linguists

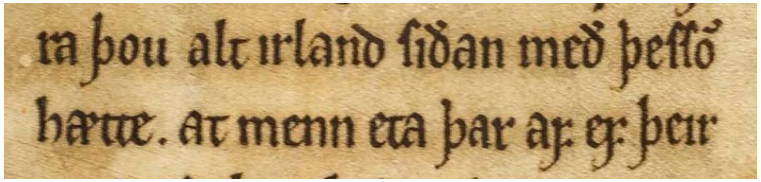
4 The Menota Lemmatizing Assistant (MLA)

Indexing Menotic XML documents

How are Menota XML texts encoded in the corpus tool?

Three words and a line beginning:

```
<w xml:id="w94400" me:msa="xAP" lemma="með">
  <choice>
    <me:facs>með</me:facs>
    <me:dipl>með</me:dipl>
    <me:norm>með</me:norm>
   </choice>
</w>
<w xml:id="w94500" me:msa="xDD gM nS cD" lemma="þessi">
  <choice>
    <me:facs>þe&slong;&slong;o<am>&bar;</am></me:facs>
    <me:dipl>þesso<ex>m</ex></me:dipl>
    <me:norm>þessum</me:norm>
  </choice>
</w>
<lb n="13" ed="ms"/>
<w xml:id="w94600" me:msa="xNC gM nS cD sI" lemma="háttr">
  <choice>
    <me:facs>hæ&trot;&trot;e</me:facs>
    <me:dipl>hætte</me:dipl>
    <me:norm>hætti</me:norm>
  </choice>
</w>
```



með þæsso*m* | hætte

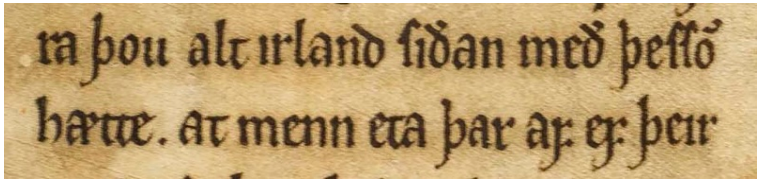
NRA 58C, 2r A.12–13

Indexing Menotic XML documents

How are Menota XML texts encoded in the corpus tool?

Three words and a line beginning:

```
<w xml:id="w94400" me:msa="xAP" lemma="með">
  <choice>
    <me:facs>með</me:facs>
    <me:dipl>með</me:dipl>
    <me:norm>með</me:norm>
   </choice>
</w>
<w xml:id="w94500" me:msa="xDD gM nS cD" lemma="þessi">
  <choice>
    <me:facs>þe&slong;&slong;o<am>&bar;</am></me:facs>
    <me:dipl>þesso<ex>m</ex></me:dipl>
    <me:norm>þessum</me:norm>
  </choice>
</w>
<lb n="13" ed="ms"/>
<w xml:id="w94600" me:msa="xNC gM nS cD sI" lemma="háttr">
  <choice>
    <me:facs>hæ&trot;&trot;e</me:facs>
    <me:dipl>hætte</me:dipl>
    <me:norm>hætti</me:norm>
  </choice>
</w>
```



með þæssom | hætte

NRA 58C, 2r A.12–13

The XML structure is transformed into tokens, each word and each structural element (XML tag) (outside of a word) being a token.

Indexing Menotic XML documents

Vertical file, consisting of one token per line, together with its attributes

```
norm                    dipl              facs              lemma     morph                  pos          xml:id   ...

...
með                     með               með               |með|     | xAP |                |prep|       w94400   ...
þessum                  þesso<ex>m</ex>   þeſſo<am>ᷣ</am>   |þessi|   | xDD gM nS cD |       |det-dem|    w94500   ...
<lb n="13" ed="ms"/>                                                                                          ...
hætti                   hætte             hærre             |háttr|   | xNC gM nS cD sI |    |noun-com|   w94600   ...
...
```

This file is compiled into internal index structures that allow for efficient querying.

From this internal representation, the XML text can be faithfully reconstructed. At the same time, the internal representation offers needed flexibility.

Text display and styling

Display of texts coded in Menotic XML can mostly be handled using XSLT stylesheets and CSS.

The basis for text display is the corpus index, not the original XML document.

Display challenges:

- Overlapping structures coded using milestones in the format <me:textSpan> … <anchor> (e.g., supplied, sic/corr)
- Pagination (as a special case of overlapping structure)

Pagination

Pagination cannot be achieved by XSLT transformations alone.

Solution:

- Build an XML fragment of the text from page beginning to next page beginning on the fly.
- This fragment is not syntactically valid because it may contain start tags without corresponding end tags and vice versa.
- Add missing start tags at start and missing end tags at end.
- The resulting XML document should be a proper sub-document of the complete one, obtainable by deleting complete XML elements (branches) from the original XML (tree).
- This ensures that the stylesheet still works properly.

Pagination: Example

XML fragment (AM 35 fol 1)

```
<pb ed="ms" n="1"/>
<lb ed="ms"/>
<div type="saga" n="Ynglinga saga">
 <div type="chapter" n="1">
  <p>
   <hi>
    <w><me:dipl>Kringla</me:dipl></w>
    <w><me:dipl>heimſins</me:dipl></w></hi>
    <w><me:dipl>ſu</me:dipl></w>
    ...
   <w><me:dipl><ex>oc</ex></me:dipl></w>
   <w><me:dipl>kulþa</me:dipl></w>
   <w><me:dipl>s<ex>vá</ex></me:dipl></w>
   <pb ed="ms" n="2"/>
```

Pagination: Example

Add minimal set of tags to make valid proper sub-document

```
<text xml:lang="isl">
 <body>
  <div n="Heimskringla" type="work">
   <pb ed="ms" n="1"/>
   <lb ed="ms"/>
   <div type="saga" n="Ynglinga saga">
    <div type="chapter" n="1">
     <p>
      <hi>
       <w><me:dipl>Kringla</me:dipl></w>
       <w><me:dipl>heimſins</me:dipl></w></hi>
       <w><me:dipl>ſu</me:dipl></w>
       ...
      <w><me:dipl><ex>oc</ex></me:dipl></w>
      <w><me:dipl>kulþa</me:dipl></w>
      <w><me:dipl>s<ex>vá</ex></me:dipl></w>
      <pb ed="ms" n="2"/>
     </p>
    </div>
   </div>
  </div>
 </body>
</text>
```

Other editorial features

- Display of up to three text levels (facs, dipl, norm) in parallel
- Scalable facsimile photographs
- Translations aligned at sentence level (one language for now; to be extended)
- Dictionary lookup – Fritzner for now; linking to ONP (Ordbog over det norrøne prosasprog) in preparation

Content

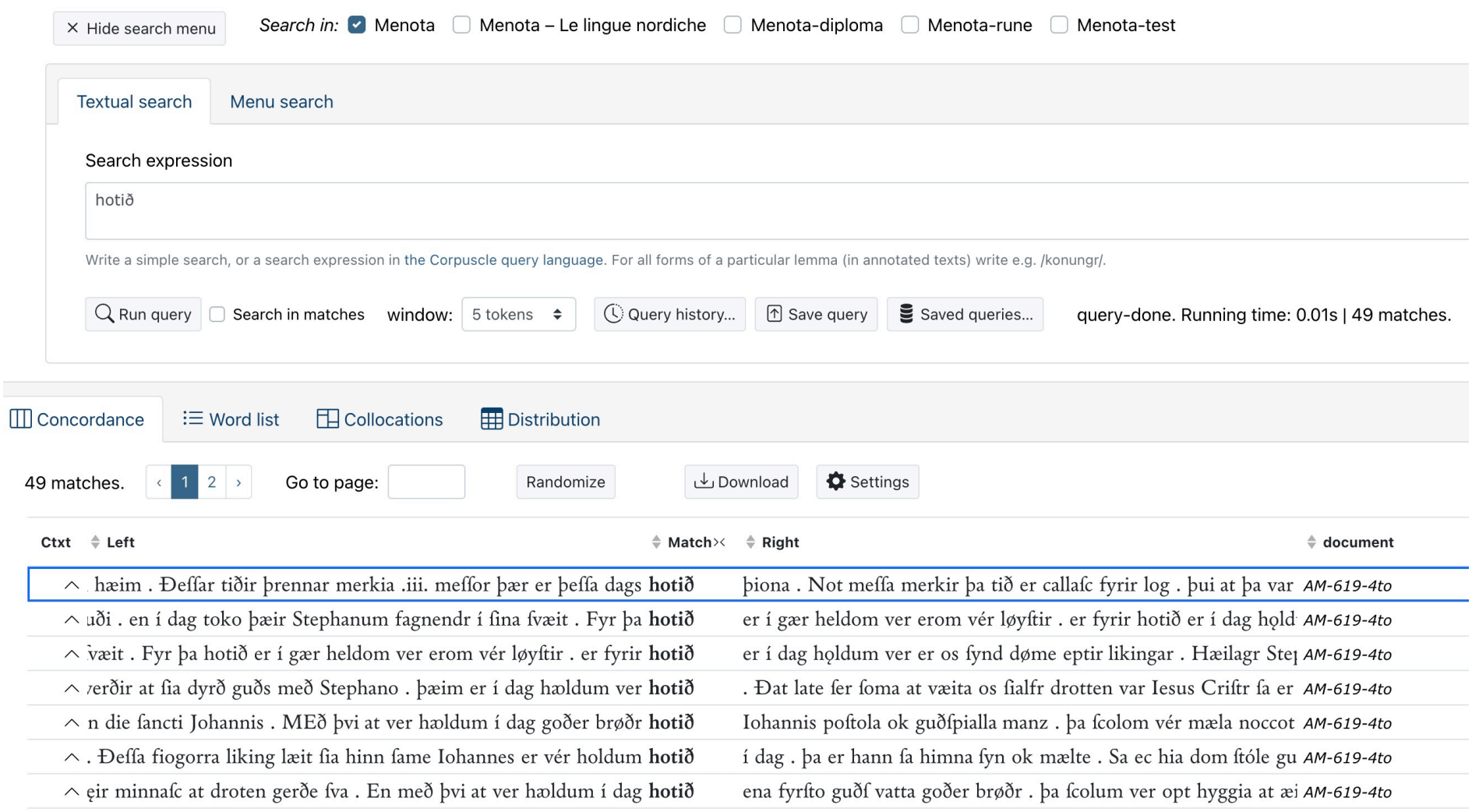
1 Overview: Menota and the Korpuskel application

2 Korpuskel as an editorial tool

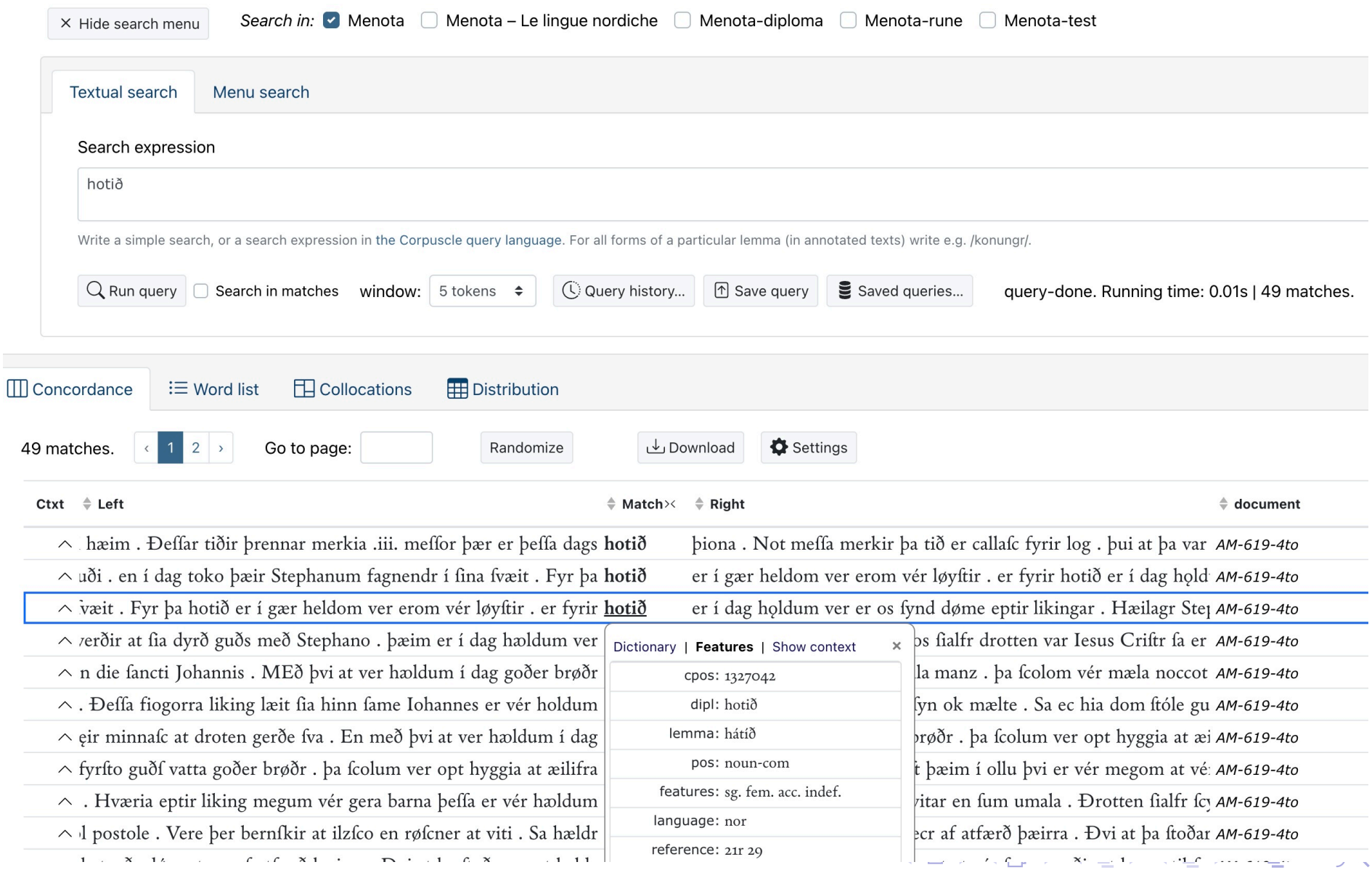
3 Korpuskel (and INESS) as tools for linguists

4 The Menota Lemmatizing Assistant (MLA)

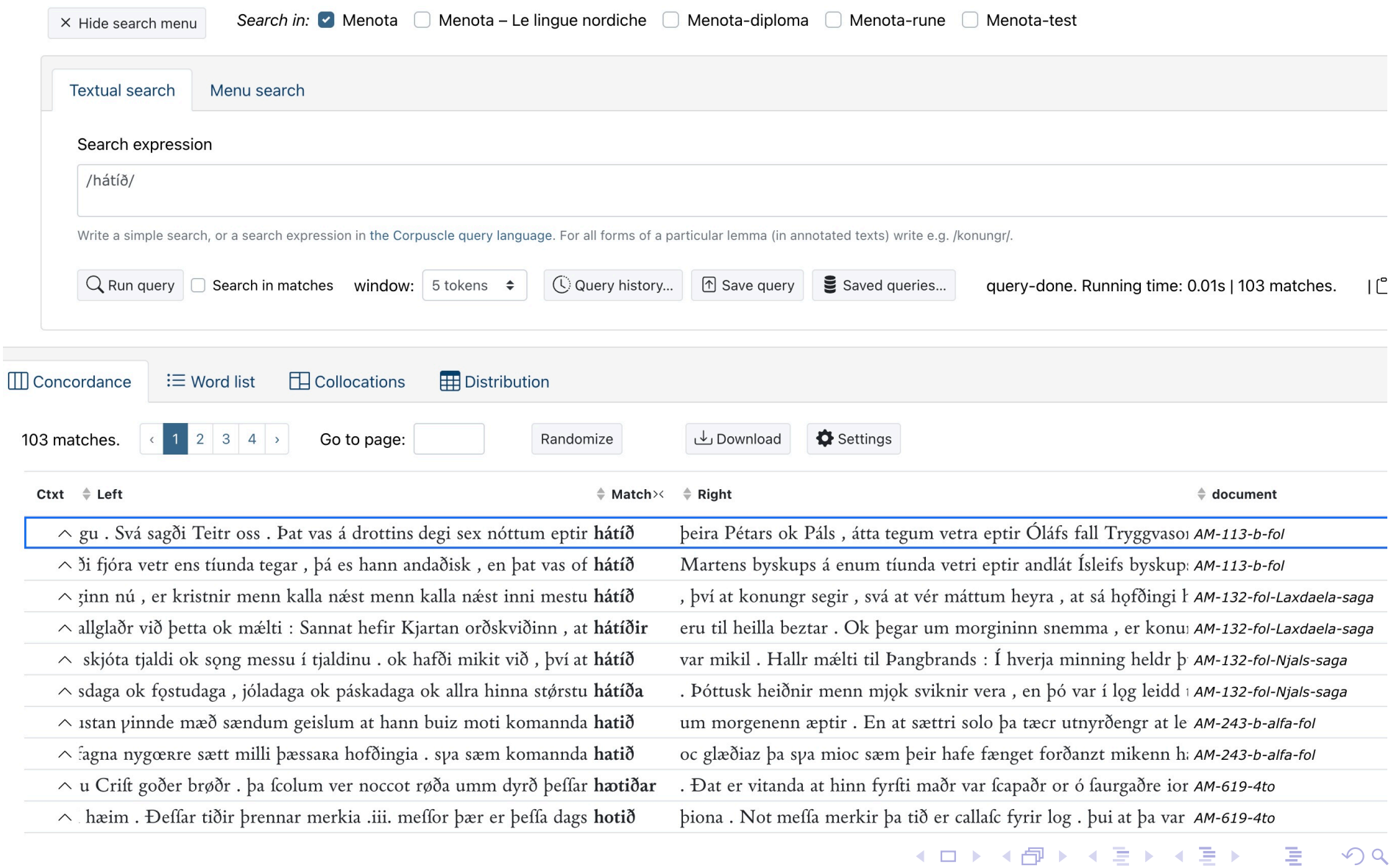
Korpuskel search: Word query example



Korpuskel search: Word query example



Korpuskel search: Lemma query example



Korpuskel search: Lemma query example



Korpuskel search: Interesting example

Allows complex queries for combinations of word, lemma, word class, morphological features, and metadata (including regular expressions)

Korpuskel search: Interesting example

Allows complex queries for combinations of word, lemma, word class, morphological features, and metadata (including regular expressions)

Example: Find attributive adjectives following the head noun

Korpuskel search: Interesting example

Allows complex queries for combinations of word, lemma, word class, morphological features, and metadata (including regular expressions)

Example: Find attributive adjectives following the head noun

Naïve search expression:

```
[pos="noun-com"] [pos="adj"] :: document='AM-619-4to'
```

Korpuskel search: Interesting example

Allows complex queries for combinations of word, lemma, word class, morphological features, and metadata (including regular expressions)

Example: Find attributive adjectives following the head noun

Naïve search expression:

```
[pos="noun-com"] [pos="adj"] :: document='AM-619-4to'
```

Old Norwegian Homily book: 330 matches

330 matches, 306 unique values. ‹ 1 2 3 4 › Go to page: Attribute: word Sort: by frequer

ignore case | Download include counts include fractions

þorf goðer	4	tið tøkeleg	2	al gørfi rétz	1
droten ſialfr	3	trv retta	2	almoſu ſlica	1
guð almatkan	3	ſtaðfeſte goðra	2	anda auðvældan	1
guð ſialfr	3	Ande heilagr	1	anda helgum	1
andanſ hælga	2	Ande hæilagr	1	andanom hælga	1

Korpuskel search: Interesting example

Search for the unmarked order, adjective preceding noun

```
[pos="adj"] [pos="noun-com"] :: document='AM-619-4to'
```

Old Norwegian Homily book: 1479 matches

1479 matches, 1175 unique values. ‹ 1 2 3 4 5 ... 15 › Go to page: Attribute: word Sort: by frequency Th

ignore case | Download include counts include fractions

goðer brøðr	27	goðer vinir	5	oarga dyr	4	hælga konungs	
goð værc	13	goðra luta	5	tiunda lut	4	hælgan anda	
hæilags anda	9	góð værc	5	veſle maðr	4	hælgum ritningum	
goðum vercum	8	hælga anda	5	ſialfr drotten	4	illi maðr	
himneſca luti	7	høgre handar	5	auma menn	3	illra manna	
hæilog ritning	7	veſla menn	5	criſtin mann	3	mykil þorf	

Korpuskel search: Limitations

Not easy to formulate agreement. We have to search for each case separately, e.g.:

```
{xNC gM nS cN}  {xAJ gM nS cN}
```

Korpuskel search: Limitations

Not easy to formulate agreement. We have to search for each case separately, e.g.:

```
{xNC gM nS cN}  {xAJ gM nS cN}
```

Even if noun and adj agree, there need not be an attributive relation

Korpuskel search: Limitations

Not easy to formulate agreement. We have to search for each case separately, e.g.:

```
{xNC gM nS cN}  {xAJ gM nS cN}
```

Even if noun and adj agree, there need not be an attributive relation:

maður	noun-com	sg. masc. nom. indef.
miſcunſamr	adj	sg. masc. nom. indef. pos.

Korpuskel search: Limitations

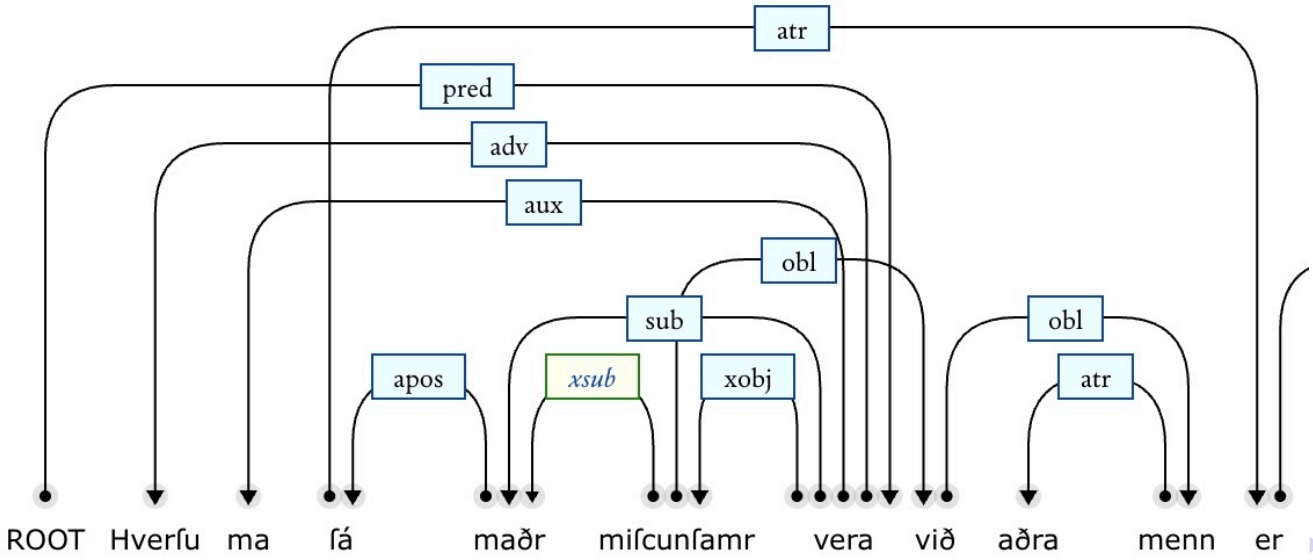
Not easy to formulate agreement. We have to search for each case separately, e.g.:

```
{xNC gM nS cN}  {xAJ gM nS cN}
```

Even if noun and adj agree, there need not be an attributive relation:

maður	noun-com	sg. masc. nom. indef.
miſcunſamr	adj	sg. masc. nom. indef. pos.

Syntactic analysis reveals the true dependency:



Syntactic search in INESS

INESS: The search conditions can be expressed exactly (in our case).

```
#n:[pos="N."] >atr #a:[pos="A-"] & #n . #a
```

Syntactic search in INESS

INESS: The search conditions can be expressed exactly (in our case).

```
#n:[pos="N."] >atr #a:[pos="A-"] & #n . #a
```

127 matches in Homily book (1312 for attribute in front of noun).

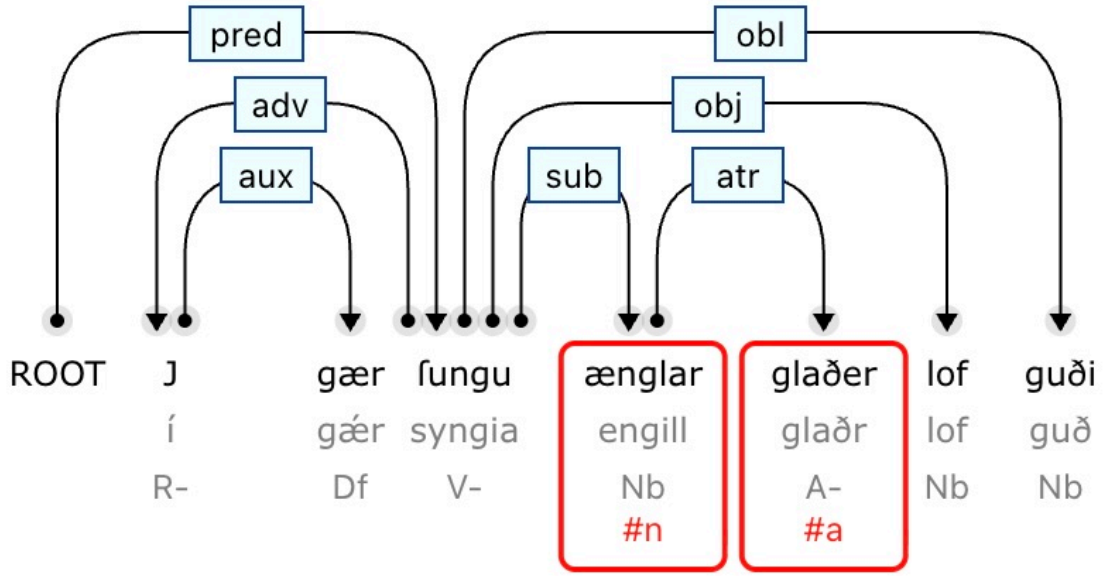
Syntactic search in INESS

INESS: The search conditions can be expressed exactly (in our case).

```
#n:[pos="N."] >atr #a:[pos="A-"] & #n . #a
```

127 matches in Homily book (1312 for attribute in front of noun).

J gær ſungu ænglar glaðer lof guði.



AM 619 4to, fol. 21r.27

Content

1 Overview: Menota and the Korpuskel application

2 Korpuskel as an editorial tool

3 Korpuskel (and INESS) as tools for linguists

4 The Menota Lemmatizing Assistant (MLA)

MLA 2.0

Features:

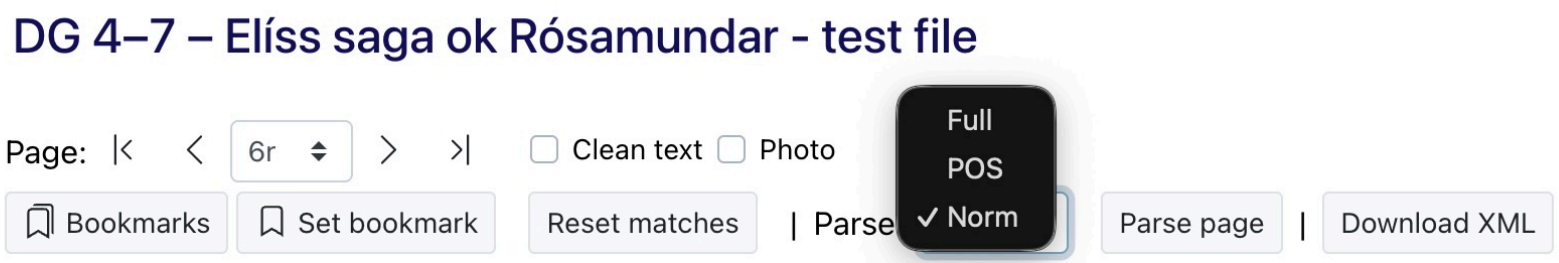
- splitting and joining tokens
- normalizing diplomatic level
- adding facsimile level
- lemmatization
- adding part of speech
- adding full morphological (+ possibly syntactic) analysis

Prerequisite: xml:id on <w> elements. The xml:id's should be consecutive, in steps of 100:

```
...   <w xml:id="w1000">…</w>
      <w xml:id="w1100">…</w>
      <w xml:id="w1200">…</w>    ...
```

This allows inserting new <w> elements, still keeping the ordering.

Choosing the annotation mode



Normalizing the diplomatic level

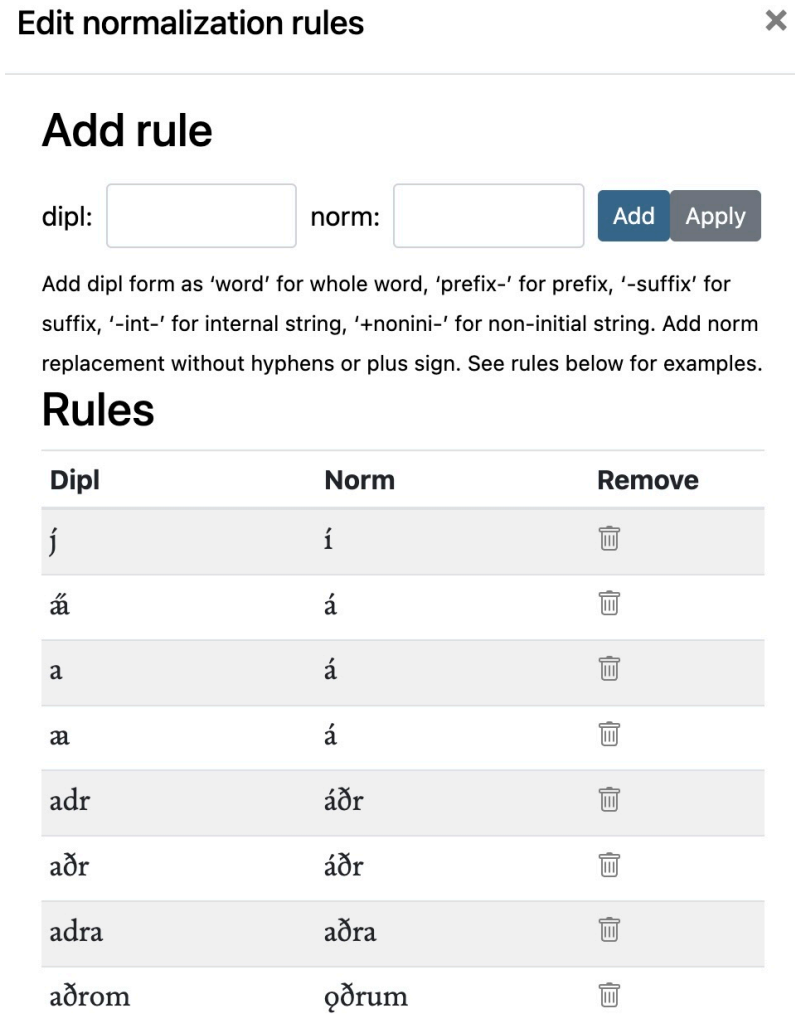
Rules that replace substrings with other substrings

Rule types:

- Full string replacement
- Prefix replacement
- Suffix replacement
- Substring replacement

More specific replacements have priority; parts of a replaced string cannot be replaced again.

Users can supply new rules.



Normalizing the diplomatic level

DG 4–7 – *6r*

‹ 1 2 3 4 5 ... 38 › Go to page:

... ‹pb/› ‹cb/› ‹head› ‹supplied/› ‹/head› ‹p› ‹lb/› HÆYRI

Dipl	Norm
‹/lb›	
hertoga	? hertoga
.	? .
er	? er
stiorn{n}	? stjǫrnn
{oc}	? ok
riki	? ríki
.	? .
valld	? vald
{oc}	? ok
‹lb n="18" ed="ms"/›	
skipan	? skipun

Edit normalization rules

ƿpp-	upp
+ɢ-	gg
+ɴ-	nn
+ʀ-	rr
+ᴛ-	tt
+ᴠ-	u
+þ-	ð
-ɑ-	a
-æ-	ǽ
-æi-	ei
-æí-	ei
-ɑʊ-	au
-azt-	ast
-c-	k
-ð-	d
-gh-	g
-ɢ-	g
-ia-	ja
-io-	jǫ

Normalizing the diplomatic level: Example

Holm perg 4 fol – Text [20]: Þiðriks saga af Bern (chs. 84–136)

Page: |< < 14r > >| Dipl Norm Translation Clean text Photo

[Dipl]

[ms: fol. 14r]

Ch. 84. her segir ym þaða risa faðyr þelents.*

Þaðe risi er a siolande synr yillcinus konongs. *ok* siokononar sem fyr þar fra sagt. at byum þæim sem faðer hans gaf hanom. Ok ekki er þæss getet at hann haþe barattomatðr* þeret nema vnat yiðr þat er hans faðer gaf hanom. þægar firir andþærðu. Vaðe risi atte son. *ok* hæitir þelent. þa er hann .ix. þættra gamall. er þaðe yill at hann neme iþrott nokkora. Spurt hæþir hann til æins smiðs. i hunalande sa hæitir mímír. *ok* er hann allra manna hagaztr. Ok þingat fær þaðe risi. mæð son sin þelænt. *ok* fæk i hond mímí. at hann skal kænna hanum iarnsmið. Siðan færr vaðe risi hæim i sioland. til bua sinna. I þænn tið þar mæð mimi Sigurðr suæinn. *ok* gerer margt illt hans smitðiusþæinum. barðe 'þa'. *ok* bæysti. Þa spurði vaðe risi at hans son þelænt var illa læikinn. firir Sigurði. *ok* gerer æptir hanom. *ok* kœmr hann hæim i sioland. Ok nu hæþir þelænt þeret i hunalande þria þætr. *ok* er hann nu .xíí. þætra gamall. Hann dþælst hæima .xíí. manaða. *ok* þockazt huærium manne þæl. *ok* allra manna er hann hagaztr. Þaðe risi spyr* or siolande huar bua tþæir dþærgar i æinu bærge er hæitir kal·

[Norm]

[ms: fol. 14r]

Ch. 84. Hér segir um Vaða risa, fǫður Velents.*

Vaði risi er á Sælandi, sonr Vilkinus konungs ok sjókonunnar, sem fyrr var frá sagt, at búum þeim sem faðir hans gaf honum. Ok ekki er þess getit at hann hafi baráttumaðr* verit, nema unat við þat, er hans faðir gaf honum, þegar fyrir ǫndverðu. Vaði risi átti sonr ok heitir Velent. Þá er hann níu vetra gamall, er Vaði vill at hann nemi íþrótt nǫkkura. Spurt hefir hann til eins smiðs í Húnalandi. Sá heitir Mímir, ok er hann allra manna hagastr. Ok þingat ferr Vaði risi með sonr sinn Velent ok fekk í hǫnd Mími at hann skal kenna honum járnsmið. Síðan ferr Vaði risi heim í Sæland til búa sinna. Í þann tíð var með Mími Sigurðr sveinn ok gerir margt illt hans smiðjusveinum, barði 'þá' ok beysti. Þá spurði Vaði risi at hans sonr Velent var illa leikinn fyrir Sigurði, ok gerir eftir honum, ok kemr hann heim í Sjóland. Ok nú hefir Velent verit í Húnalandi þrjá vetr, ok er hann nú tolf vetra gamall. Hann dvelsk heima tolf mánaða ok þokkask hverjum manni vel, ok allra manna er hann hagastr. Vaði risi spyrr* ór Sjólandi hvar búa tveir dvergar í einu bergi er heitir Kal·

lava. Þessir dvergar kunnu betr smíða af járni en engi aðrir,

Annotating lemma, part of speech and features

All readings found in the corpus are shown, ordered by frequency

Dipl		MSA	
<lb n="2" ed="ms"/>			
sínum	3	sinn xDP cD nP gU	[6] ...
<lb n="3" ed="DN"/>			
þæím	2	sá xDD cD nP gU	[21] ...
sem	9	sem xCS	[665] ...
þetta	6	sjá xDD cA nS gN	[720]
		sjá xDD cN nS gN	[471]
		sjá xPD cA nS gN	[62]
		sjá xPD cN nS gN	[29]
		þessi xDD cA nS gN	[8]
		þessi xDD cN nS gN	[5]
bræf	4	bréf xNC cA nP gN sI	[23] ...

Annotating lemma, part of speech and features

The user choses the correct reading

Dipl		MSA
‹lb n="2" ed="ms"/›		
sínum	3	sinn xDP cD nP gU [6] ...
‹lb n="3" ed="DN"/›		
þæím	2	sá xDD cD nP gU [21] ...
sem	9	sem xCS [665] ...
þetta	1/6	✓✓ sjá xDD cA nS gN [720]
bræf	4	bréf xNC cA nP gN sI [23] ...
sia	12	sjá xVB fI tPS vA iU [69] ...
eða	2	eða xCC [1270] ...
hœyra	4	heyra xVB fI tPS vA [9] ...

Annotating lemma, part of speech and features

Doing lemmatization and morphological annotation from scratch

Dipl XML: **sia**

Norm XML: sjá ☐ empty norm

Candidates: sjá

Facs XML:

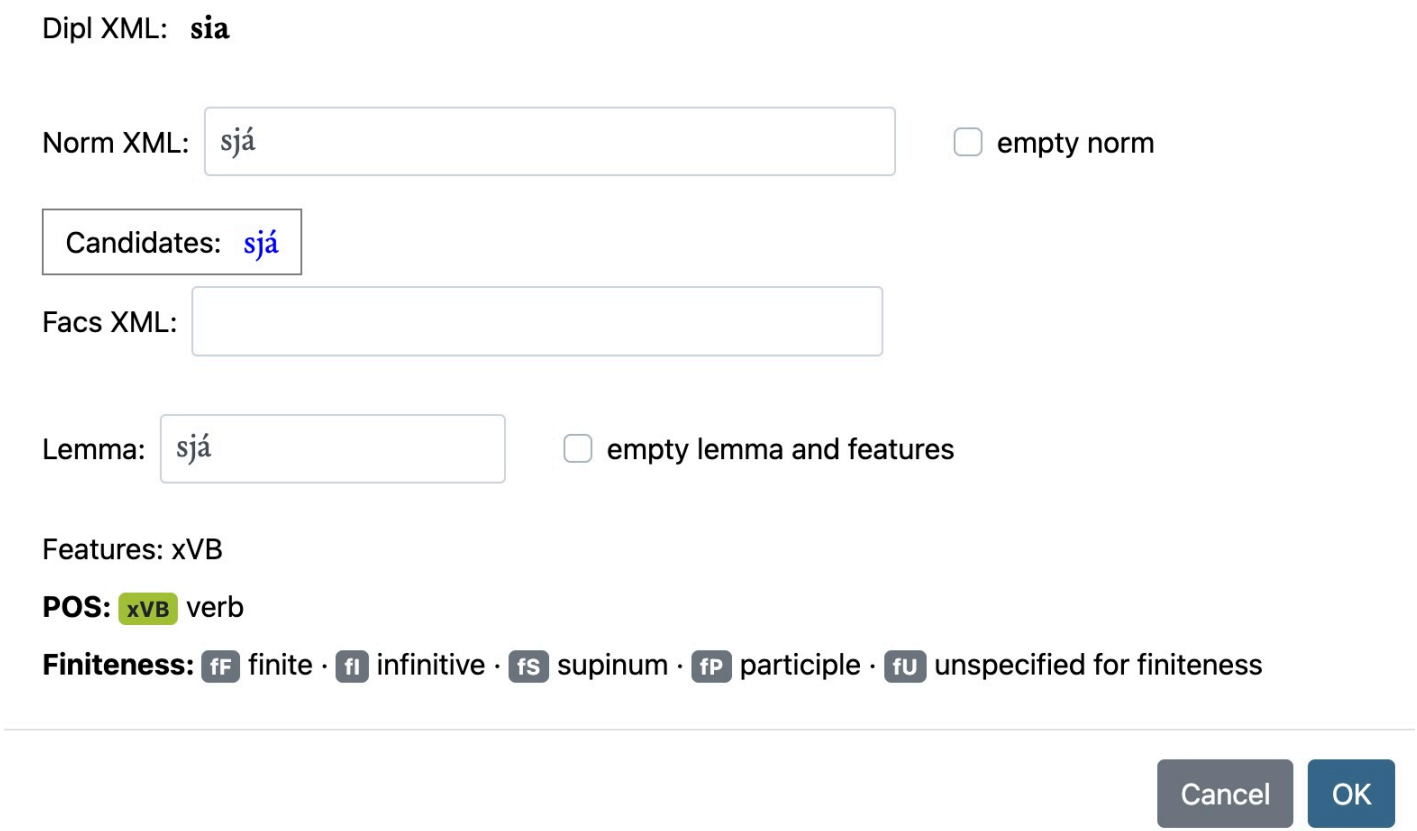
Lemma: sjá ☐ empty lemma and features

Features:

POS: xVB verb · xNC noun, common · xNP noun, proper · xAJ adjective · xAV adverb, general · xAQ adverb, interrogative · xAP preposition · xCC conjunction · xCS subjunction · xRP relative pronoun · xPE pronoun, personal · xPR pronoun, reflexive · xPQ pronoun, interrogative · xPI pronoun, indefinite · xDD determiner, demonstrative · xDQ determiner, quantifier · xDP determiner, possessive · xNA numeral, cardinal · xNO numeral, ordinal · xIM infinitive marker · xVP verbal particle · xIT interjection · xPD pronoun, general · xNU numeral, general · xCU conjunction, general · xAT article · xNE enclitic · xEX expletive particle · xFW foreign word · xFN foreign name · xUA unassigned · xEY empty

Cancel OK

Annotating lemma, part of speech and features



Annotating lemma, part of speech and features

Dipl XML: **sia**

Norm XML: sjá ☐ empty norm

Candidates: sjá

Facs XML:

Lemma: sjá ☐ empty lemma and features

Features: xVB fI

POS: xVB verb

Finiteness: fI infinitive

Tense: tPS present · tPT preterite · tU unspecified tense

Voice: vA active · vR reflexive · vP passive · vD deponent · vU unspecified voice

Cancel OK

Annotating lemma, part of speech and features

Dipl: **sia**

Dipl XML: **sia**

Norm XML: sjá ☐ empty norm

Candidates: sjá

Facs XML:

Lemma: sjá ☐ empty lemma and features

Features: xVB fI tPS

POS: xVB verb

Finiteness: fI infinitive

Tense: tPS present

Voice: vA active · vR reflexive · vP passive · vD deponent · vU unspecified voice

Cancel OK

Annotating lemma, part of speech and features

Add reading

... ‹pb/› ‹lb/› ‹lb/› ‹lb› ‹/lb› ‹seg› ‹/seg› ‹lb/› sínum ‹lb/› þæim sem þetta bræf **sia** eða hœyra Q{ueðiu} G{uðs} {ok} sína . v{er} vilíum ollum ‹lb/› mon{n}om kunnikt gera at ‹lb/› ‹lb/› ‹lb/› ‹lb/› ...

Dipl: **sia**

Dipl XML: **sia**

Norm XML: sjá ☐ empty norm

Candidates: sjá

Facs XML:

Lemma: sjá ☐ empty lemma and features

Features: xVB fI tPS vA iU

POS: xVB verb

Finiteness: fI infinitive

Tense: tPS present

Voice: vA active

Rule-based annotation?

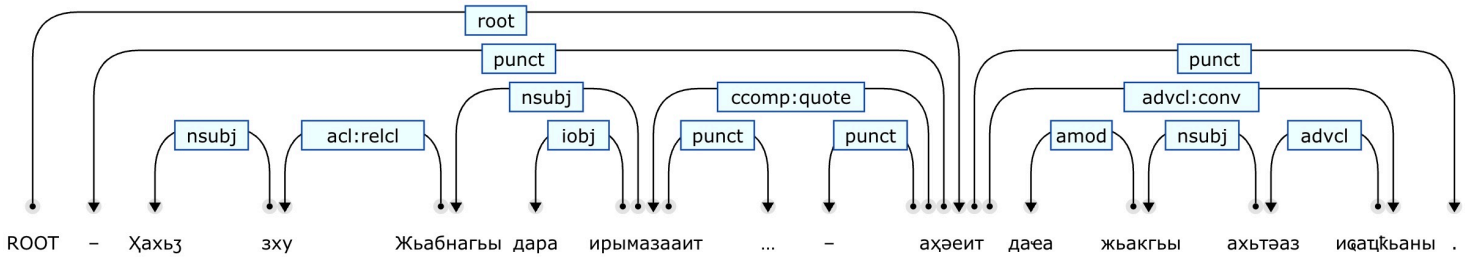
Future possible extensions:

- Context-driven disambiguation using a rule-based grammar (Constraint Grammar)
- Rule-based syntactic analysis in a dependency grammar framework with manual correction

Rule-based annotation: The AbNC example

дара/w292400	1/10	6→5	дара́ Pron Pers 3Pl <Lemma> >IOBJ *select:2186 with:5857 map:5860 setparent:5861*
ирымазааит/w292500	1	9→6	а́-ма-заа-ра V Stat Fin Subj S:3 IO:3Pl >CCOMP:QUOTE *with:8297 setparent:8302 map:8303*
.../w292600	1	6→7	... Punct Ellipsis >PUNCT *with:8029 setparent:8032 map:8033*
—/w292700	1	9→8	— Punct Dash >PUNCT *with:8880 setparent:8884 map:8885*
аҳәеит/w292800	1/2	✓ 9	а-ҳәа-ра́ V Dyn Tr Fin Aor S:3SgNH DO:3 DO:Ad >ROOT *select:1777 map:8956 setchild:9062*
			á-ҳәа-ра V Dyn Intr Fin Aor S:3 S:Ad IO:3SgNH *; select:1777*

Store



Store Remove

